

Project Initialization and Planning Phase

Date	25 June 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement):

The Voice-Based Concept Understanding Analyzer is designed to help students improve their conceptual understanding through AI-based speech analysis. Many students struggle to determine whether their spoken explanations are accurate and complete, while teachers spend significant time evaluating each student's performance manually. This project addresses these challenges by converting speech to text, analyzing conceptual understanding using AI, generating performance scores, and providing instant feedback. The following customer problem statements represent the key challenges faced by the target users and highlight the need for an automated evaluation system.

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A student preparing for exams or viva	Explain concepts clearly and improve my understanding	I don't know whether my explanation is correct	There is no instant feedback on my spoken explanation	Confused and lacking confidence
PS-2	A teacher	Evaluate students' conceptual understanding quickly	Manual evaluation takes too much time	Each student must be assessed individually	Overloaded and tired
PS-3	An online learner	Practice speaking and improve communication skills	I don't receive personalized feedback	Most platforms only evaluate quizzes, not spoken explanations	Less confident in interviews and presentations